

Xingfu Yang, PhD

📍 U.S. Permanent Resident | 📞 303 523 8836 | ✉️ xingfu@xyang.me | 🌐 xyang.me | 📍 LA, CA | 🌐 | 🌐

HIGHLIGHTS

- **Founding Data Scientist** who builds from zero — production systems, experimentation frameworks, evaluation pipelines, and the teams around them. Built data science from scratch at two startups, turning ambiguous problems into measurable outcomes.
- PhD-trained **applied scientist**: scoring, ranking, classification, fraud detection, personalization, and GenAI at scale (40M+ entities, billions of records). Full ML lifecycle from research to production monitoring.
- **Research depth meets production pragmatism**: trained a foundation model from scratch on TPUs, scaled GNNs to 1.6B edges, completed advanced coursework from Stanford and Berkeley. 6 years balancing scientific rigor with building systems that ship.

WORK EXPERIENCE

Lucid Intel

Santa Monica, CA

Chief Data Scientist

Jun 2023 – Feb 2026

- Built the data science function from zero: defined metrics, designed pipelines, built evaluation frameworks, and delivered actionable insights that shaped product roadmap and business strategy.
- Designed, trained, and deployed scoring, classification, and fraud detection models on 40M+ identities and billions of records — supervised and unsupervised deep learning with rigorous evaluation and production monitoring.
- Led A/B experimentation programs: designed controlled experiments, conducted causal analysis, and translated complex results into clear, data-driven recommendations for cross-functional teams.
- Built LLM-based production systems — agentic workflows, RAG pipelines, and generative applications. Developed evaluation benchmarks to measure quality and identify failure modes.

Leap Theory

Los Angeles, CA

Data Scientist

Oct 2019 – Jun 2023

- Built production ML systems from scratch: scoring, classification, and recommendation models with tiered caching for high-throughput, low-latency serving.
- Achieved a 55% reduction in noise, 70% increase in accept ratio, and 2–10 pp improvement in first-payment default through rigorous feature engineering and model iteration.

INDEPENDENT PROJECTS

MiniLM: Foundation Model from Scratch (JAX / Google TPU Research Cloud)

- Multi-device sharded pretraining of a Mixture-of-Experts foundation model using the OpenWebText dataset with data and tensor parallelism.
- Post-training: supervised fine-tuning, reward modeling, and RLHF alignment via PPO, DPO, and GRPO.

Scaling GNNs to Billion-Edge Graphs (PyTorch / PyG)

- Trained GraphSAGE on a 100 million node, 1.6 billion edge graph — scalable representation learning and pattern discovery on large-scale relational data.

EXPERTISE

Applied ML: Scoring, Ranking, Classification, Fraud Detection, Personalization, A/B Testing, Causal Inference, Evaluation Frameworks

ML Frameworks: PyTorch, TensorFlow, JAX/FLAX, Triton, Keras, XGBoost, Scikit-Learn, AutoGluon, HuggingFace

GenAI & LLMs: Transformers, RLHF (PPO, DPO, GRPO), RAG, Agentic AI, LangChain, LangGraph, Fine-Tuning (LoRA)

Languages: Python (7+ yrs), SQL (7+ yrs), C/C++ (CUDA, MPI, OpenMP), R, Bash

Python Stack: FastAPI, Django, Flask, Gunicorn, SQLAlchemy, NumPy, Numba, Dask, Pandas, Matplotlib, Seaborn

Orchestration: Kubernetes, Ray, Slurm, Kubeflow, Airflow, Docker, TFX, TensorFlow Serving, Prometheus, Grafana

Cloud: AWS Solutions Architect Associate, GCP (GKE, TPU Research Cloud, BigQuery, Vertex AI)

Coursework: Language Modeling from Scratch (Stanford CS336), Deep Generative Models (Stanford CS236), Deep Learning (Berkeley CS182), Deep Reinforcement Learning (Berkeley CS285), ML with Graphs (Stanford CS224W)

RESEARCH EXPERIENCE

Research Assistant, Colorado School of Mines

Golden, CO

PhD in Chemical Engineering (Computational Physics); GPA: 3.86/4.00

Aug 2013 – Aug 2019

- Developed numerical simulation of patterning of non-spherical colloids under electric fields. Published in *Physical Review Letters*.
- Teamed with collaborators to design an $O(N_b \log N_b)$ framework to model a large number of colloids.
- Simulated in- and out-of-equilibrium behaviors of particles in a high-performance computing cluster.